Invariants of Conic Sections

Amer Rawashdeh

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Move an ellipse in the plane, by turning it or sliding it, and its equation changes completely, yet it is the same curve. This project answers the natural question: which numbers, computed from the equation, do *not* change? Two such numbers turn out to be enough. Two conics with the same two numbers are the same curve in different positions.

1 The setting

A *conic* is the set of points (x, y) in the plane where a quadratic polynomial vanishes:

$$p(x,y) = ax^2 + 2bxy + cy^2 + 2dx + 2ey + f = 0, \qquad (a,b,c) \neq (0,0,0). \tag{1}$$

The factors 2 are there so that the coefficients fit into a symmetric matrix. With $v = (x, y, 1)^T$,

$$p(x,y) = v^T M v, \qquad M = \begin{pmatrix} a & b & d \\ b & c & e \\ d & e & f \end{pmatrix}, \qquad M_0 = \begin{pmatrix} a & b \\ b & c \end{pmatrix},$$

where M_0, the top-left corner of M, holds the quadratic terms alone.

> Calculation
>
> Multiply out:
>
> $$v^T M v = \begin{pmatrix} x & y & 1 \end{pmatrix} \begin{pmatrix} ax+by+d \\ bx+cy+e \\ dx+ey+f \end{pmatrix} = ax^2+bxy+dx+bxy+cy^2+ey+dx+ey+f = p(x,y).$$

Three numbers are made from the matrix:

$$T = a + c = \operatorname{tr} M_0, \qquad \delta = ac - b^2 = \det M_0, \qquad \Delta = \det M.$$

An equation up to a factor. The equations $p = 0$ and $\lambda p = 0$, for a number $\lambda \neq 0$, describe the same conic. So a number that truly belongs to the conic, and not to one particular equation of it, must not change when p is replaced by λp. Throughout, "a conic" means the equation $p = 0$ up to such a factor.[1]

[1] For a non-degenerate conic with real points this is the same as the set of points, since the set determines the equation up to a factor. For an equation like $x^2 + y^2 + 1 = 0$, with no real points at all, the equation carries more information than the (empty) set.

2 The main result

Theorem 1. *Suppose $\Delta \neq 0$. The two numbers*

$$f_1 = \frac{a+c}{\Delta^{1/3}} = \frac{T}{\Delta^{1/3}}, \qquad f_2 = \frac{ac-b^2}{\Delta^{2/3}} = \frac{\delta}{\Delta^{2/3}}$$

(real cube roots) do not change when the equation is multiplied by a nonzero number, nor when the conic is moved by a rigid motion: a rotation, a reflection, a translation, or any combination of them.

The proof has two independent parts:

- a rigid motion does not change T, δ or Δ at all (Section 3);
- multiplying p by λ multiplies T, δ, Δ by λ, λ^2, λ^3, and the powers of Δ in f_1, f_2 are exactly the ones that cancel these factors (Section 4).

Section 5 shows that f_1 and f_2 *do* change when the conic is enlarged, so they see the size of a conic, not only its shape. Sections 6 and 7 explain where Δ comes from and what the condition $\Delta \neq 0$ means, and Section 8 shows that f_1 and f_2 are enough to tell any two conics apart.

3 Rigid motions

A *rigid motion* of the plane is a map

$$\begin{pmatrix} x \\ y \end{pmatrix} \longmapsto Q \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} t_1 \\ t_2 \end{pmatrix}, \qquad Q^T Q = I.$$

The matrix Q is *orthogonal*: it keeps lengths and angles. The orthogonal 2×2 matrices are the rotations and the reflections,

$$Q_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \ (\det Q_\theta = 1), \qquad \begin{pmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{pmatrix} \ (\det = -1).$$

The rotation Q_θ is multiplication by the complex number $\cos\theta + i\sin\theta$, of absolute value 1, written as a matrix.

Calculation

Write $\beta = \cos\theta$, $\gamma = \sin\theta$, so $\beta^2 + \gamma^2 = 1$. Then

$$(\beta + i\gamma)(x + iy) = (\beta x - \gamma y) + i(\gamma x + \beta y), \quad \text{that is,} \quad \begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} \beta & -\gamma \\ \gamma & \beta \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}.$$

Multiplying by a complex number of absolute value 1 turns the plane without stretching it.

The key observation. With the extra coordinate 1, a rigid motion is a single matrix product $v \mapsto Gv$:

$$G = \begin{pmatrix} Q & t \\ 0\ 0 & 1 \end{pmatrix} = \begin{pmatrix} q_{11} & q_{12} & t_1 \\ q_{21} & q_{22} & t_2 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \det G = \det Q = \pm 1.$$

Substituting the moved point into the equation gives

$$p(Gv) = (Gv)^T M(Gv) = v^T \big(G^T M G\big) v,$$

so **the new coefficient matrix is** $M' = G^T M G$. The matrix of the motion appears on the right, and its transpose on the left. (For a rotation the transpose is also the inverse, $Q_\theta^T = Q_{-\theta}$, which is why the left factor looks like the inverse rotation.)

Proposition 2. *A rigid motion does not change T, δ and Δ.*

Proof. For Δ: $\det M' = \det(G^T)\det M \det G = (\det G)^2 \Delta = \Delta$, because $\det G = \pm 1$.

For T and δ: the top-left corner of $G^T M G$ is $M_0' = Q^T M_0 Q$, whatever the translation t is. Since $Q^T = Q^{-1}$, the matrix $M_0' = Q^{-1} M_0 Q$ is similar to M_0, and similar matrices have the same trace and the same determinant. □

Calculation

Write M and G in blocks, with $w = (d, e)^T$:

$$M = \begin{pmatrix} M_0 & w \\ w^T & f \end{pmatrix}, \qquad G = \begin{pmatrix} Q & t \\ 0 & 1 \end{pmatrix}, \qquad G^T = \begin{pmatrix} Q^T & 0 \\ t^T & 1 \end{pmatrix}.$$

Then

$$MG = \begin{pmatrix} M_0 Q & M_0 t + w \\ w^T Q & w^T t + f \end{pmatrix}, \qquad G^T M G = \begin{pmatrix} Q^T M_0 Q & Q^T(M_0 t + w) \\ * & * \end{pmatrix}.$$

The corner contains no t: a translation never changes the quadratic terms.

The two basic motions, written out coefficient by coefficient, follow. The primes denote the coefficients of the new equation $p'(x, y) = p(Gv) = a'x^2 + 2b'xy + c'y^2 + 2d'x + 2e'y + f'$.

A rotation. Substituting $x \mapsto \beta x - \gamma y$, $y \mapsto \gamma x + \beta y$ ($\beta = \cos\theta$, $\gamma = \sin\theta$):

$$\begin{aligned} a' &= a\beta^2 + 2b\beta\gamma + c\gamma^2, & b' &= (c-a)\beta\gamma + b(\beta^2 - \gamma^2), & c' &= a\gamma^2 - 2b\beta\gamma + c\beta^2, \\ d' &= \beta d + \gamma e, & e' &= \beta e - \gamma d, & f' &= f. \end{aligned}$$

Calculation

Expand $p(\beta x - \gamma y,\ \gamma x + \beta y)$ and collect terms. The quadratic part:

$$\begin{aligned} a(\beta x - \gamma y)^2 &= a\big(\beta^2 x^2 - 2\beta\gamma\, xy + \gamma^2 y^2\big), \\ 2b(\beta x - \gamma y)(\gamma x + \beta y) &= 2b\big(\beta\gamma\, x^2 + (\beta^2 - \gamma^2)\, xy - \beta\gamma\, y^2\big), \\ c(\gamma x + \beta y)^2 &= c\big(\gamma^2 x^2 + 2\beta\gamma\, xy + \beta^2 y^2\big). \end{aligned}$$

The coefficient of x^2 is $a\beta^2 + 2b\beta\gamma + c\gamma^2$, of y^2 is $a\gamma^2 - 2b\beta\gamma + c\beta^2$, and of xy is $2\big[(c - a)\beta\gamma + b(\beta^2 - \gamma^2)\big]$. The linear part:

$$2d(\beta x - \gamma y) + 2e(\gamma x + \beta y) = 2(\beta d + \gamma e)\, x + 2(\beta e - \gamma d)\, y.$$

A direct check: $a' + c' = (a + c)(\beta^2 + \gamma^2) = a + c$. In matrix form the whole substitution is

$$\begin{pmatrix} a' & b' & d' \\ b' & c' & e' \\ d' & e' & f' \end{pmatrix} = \begin{pmatrix} \beta & \gamma & 0 \\ -\gamma & \beta & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} a & b & d \\ b & c & e \\ d & e & f \end{pmatrix} \begin{pmatrix} \beta & -\gamma & 0 \\ \gamma & \beta & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

with both outer matrices of determinant $\beta^2 + \gamma^2 = 1$.

A translation. Substituting $x \mapsto x + t_1$, $y \mapsto y + t_2$ leaves a, b, c unchanged and gives

$$d' = at_1 + bt_2 + d, \qquad e' = bt_1 + ct_2 + e, \qquad f' = p(t_1, t_2).$$

Calculation

Expand:

$$\begin{aligned} p(x + t_1, y + t_2) &= a(x + t_1)^2 + 2b(x + t_1)(y + t_2) + c(y + t_2)^2 + 2d(x + t_1) + 2e(y + t_2) + f \\ &= ax^2 + 2bxy + cy^2 + 2(at_1 + bt_2 + d)\, x + 2(bt_1 + ct_2 + e)\, y + p(t_1, t_2). \end{aligned}$$

In matrix form,

$$\begin{pmatrix} a' & b' & d' \\ b' & c' & e' \\ d' & e' & f' \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ t_1 & t_2 & 1 \end{pmatrix} \begin{pmatrix} a & b & d \\ b & c & e \\ d & e & f \end{pmatrix} \begin{pmatrix} 1 & 0 & t_1 \\ 0 & 1 & t_2 \\ 0 & 0 & 1 \end{pmatrix},$$

and both outer matrices have determinant 1, so $\Delta' = \Delta$. Since a, b, c do not change, neither do T and δ.

4 Rescaling the equation

Replacing p by λp replaces M by λM, so

$$T \mapsto \lambda T, \qquad \delta \mapsto \lambda^2 \delta, \qquad \Delta \mapsto \lambda^3 \Delta.$$

The reason is degree: T is of degree 1 in the coefficients, δ of degree 2, Δ of degree 3. Dividing T by $\Delta^{1/3}$ and δ by $\Delta^{2/3}$ gives expressions of degree 0, and these are exactly f_1 and f_2. This completes the proof of Theorem 1.

Calculation

$\det(\lambda M) = \lambda^3 \det M$ for a 3×3 matrix, and $\det(\lambda M_0) = \lambda^2 \det M_0$. Then

$$\frac{\lambda T}{(\lambda^3 \Delta)^{1/3}} = \frac{\lambda T}{\lambda\, \Delta^{1/3}} = f_1, \qquad \frac{\lambda^2 \delta}{(\lambda^3 \Delta)^{2/3}} = \frac{\lambda^2 \delta}{\lambda^2\, \Delta^{2/3}} = f_2.$$

The real cube root satisfies $(\lambda^3 \Delta)^{1/3} = \lambda\, \Delta^{1/3}$ also for negative λ, which is why real cube roots are used.

Remark (why Δ is needed). The quotient $T^2/\delta = (a+c)^2/(ac-b^2)$ is also unchanged by rigid motions and by rescaling, and it does not involve Δ. But it ignores the linear and constant terms completely: every circle gives the value 4, whatever its radius, and every ellipse $x^2/\alpha^2 + y^2/\beta^2 = 1$ gives $(\alpha^2+\beta^2)^2/(\alpha\beta)^2$, which depends only on the ratio α/β. It sees the shape of a conic and not its size. In fact $T^2/\delta = f_1^2/f_2$: it is the combination of f_1 and f_2 that forgets the size (see the next section).

5 Enlarging is not a rigid motion

Enlarging the plane by a factor $\mu \neq 0$, that is substituting $x \mapsto \mu x$, $y \mapsto \mu y$, multiplies the quadratic terms by μ^2, the linear terms by μ, and leaves the constant alone. Then

$$T \mapsto \mu^2 T, \qquad \delta \mapsto \mu^4 \delta, \qquad \Delta \mapsto \mu^4 \Delta, \qquad \text{so} \qquad f_1 \mapsto \mu^{2/3} f_1, \qquad f_2 \mapsto \mu^{4/3} f_2.$$

So f_1 and f_2 change (unless $\mu = \pm 1$, and $\mu = -1$ is the rotation by π). They measure size as well as shape. For example, for the circle $x^2 + y^2 = r^2$,

$$f_1 = -2\,r^{-2/3}, \qquad f_2 = r^{-4/3},$$

so circles of different radii have different values.

Calculation

The new matrix is SMS with $S = \operatorname{diag}(\mu, \mu, 1)$:

$$\begin{pmatrix} \mu^2 a & \mu^2 b & \mu d \\ \mu^2 b & \mu^2 c & \mu e \\ \mu d & \mu e & f \end{pmatrix} = \begin{pmatrix} \mu & 0 & 0 \\ 0 & \mu & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} a & b & d \\ b & c & e \\ d & e & f \end{pmatrix} \begin{pmatrix} \mu & 0 & 0 \\ 0 & \mu & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

so $\Delta \mapsto (\det S)^2 \Delta = \mu^4 \Delta$. Then

$$f_1 \mapsto \frac{\mu^2 T}{\mu^{4/3} \Delta^{1/3}} = \mu^{2/3} f_1, \qquad f_2 \mapsto \frac{\mu^4 \delta}{\mu^{8/3} \Delta^{2/3}} = \mu^{4/3} f_2.$$

For the circle, $p = x^2 + y^2 - r^2$, so $a = c = 1$, $b = d = e = 0$, $f = -r^2$, and $T = 2$, $\delta = 1$, $\Delta = -r^2$.

6 Where Δ comes from: the center

When $\delta \neq 0$, the conic has a *center*: the point (x_0, y_0) where both partial derivatives of p vanish,

$$ax_0 + by_0 + d = 0, \qquad bx_0 + cy_0 + e = 0, \qquad \text{so} \qquad x_0 = \frac{be - cd}{\delta}, \quad y_0 = \frac{bd - ae}{\delta}.$$

Translating the center to the origin removes the linear terms (these are the formulas for d', e' in Section 3), and the constant term becomes the value of p at the center:

Proposition 3. *If $\delta \neq 0$, then $p(x_0, y_0) = \dfrac{\Delta}{\delta}$, that is, $\Delta = \delta \cdot p(\text{center})$.*

This is where Δ comes from. A rotation then turns the axes of the conic to the coordinate axes (the spectral theorem: the symmetric matrix M_0 is $Q^T \operatorname{diag}(\lambda_1, \lambda_2) Q$ for a rotation Q), and the equation becomes

$$\lambda_1 x^2 + \lambda_2 y^2 + \frac{\Delta}{\delta} = 0, \qquad \lambda_1 + \lambda_2 = T, \quad \lambda_1 \lambda_2 = \delta, \tag{2}$$

where λ_1, λ_2 are the eigenvalues of M_0.

Calculation

Completing the square, first in x and then in y, shows the same thing when $a \neq 0$:

$$\begin{aligned} p(x,y) &= a\Big(x^2 + \tfrac{2b}{a}xy\Big) + cy^2 + 2dx + 2ey + f \\ &= a\Big(x + \tfrac{b}{a}y\Big)^2 + \tfrac{ac-b^2}{a}\,y^2 + 2d\Big(x + \tfrac{b}{a}y\Big) + 2\,\tfrac{ae-bd}{a}\,y + f \\ &= a\Big(x + \tfrac{b}{a}y + \tfrac{d}{a}\Big)^2 + \tfrac{\delta}{a}\,y^2 + 2\,\tfrac{ae-bd}{a}\,y + f - \tfrac{d^2}{a} \\ &= a\Big(x + \tfrac{b}{a}y + \tfrac{d}{a}\Big)^2 + \tfrac{\delta}{a}\Big(y + \tfrac{ae-bd}{\delta}\Big)^2 + f - \tfrac{d^2}{a} - \tfrac{(ae-bd)^2}{a\,\delta}. \end{aligned}$$

The constant at the end is Δ/δ: over the common denominator $a\delta$, its numerator is

$$af\delta - d^2\delta - (ae - bd)^2 = a\big(acf - ae^2 - b^2 f + 2bde - cd^2\big) = a\,\Delta,$$

using the expansion $\Delta = acf - ae^2 - b^2 f + 2bde - cd^2$ of the 3×3 determinant. Both squares vanish exactly at the center, so the constant is $p(x_0, y_0)$.

The parabolic case $\delta = 0$. Then $T \neq 0$ (if $ac = b^2$ and $a + c = 0$, then $a = b = c = 0$), the eigenvalues of M_0 are 0 and T, and there is no center. If moreover $\Delta \neq 0$, a rotation and a translation bring the equation to

$$Ty^2 + 2kx = 0, \qquad k^2 = -\frac{\Delta}{T}. \tag{3}$$

Calculation

After a rotation, $M_0 = \operatorname{diag}(0, T)$, so $p = Ty^2 + 2d'x + 2e'y + f'$, and

$$\Delta = \det \begin{pmatrix} 0 & 0 & d' \\ 0 & T & e' \\ d' & e' & f' \end{pmatrix} = -Td'^2.$$

So $\Delta \neq 0$ means $d' \neq 0$. Completing the square in y gives $T(y + e'/T)^2 + 2d'x + (\text{constant})$, and a translation in x and in y removes both the y term and the constant. The determinant is unchanged, so $k = d'$ satisfies $\Delta = -Tk^2$.

7 The meaning of $\Delta = 0$

Proposition 4. *If p is a product of two linear polynomials, so that the conic is a pair of lines, then $\Delta = 0$.*

Proof. Write $p(x,y) = (u \cdot v)(w \cdot v)$ with $v = (x,y,1)^T$ and two fixed vectors u, w. Then $p = v^T u\, w^T v = v^T\big(\frac{1}{2}(uw^T + wu^T)\big)v$, so $M = \frac{1}{2}(uw^T + wu^T)$. Every column of this matrix is a combination of u and w, so its rank is at most 2 and $\det M = 0$. □

Calculation

With $u = (\alpha_1, \beta_1, \gamma_1)$ and $w = (\alpha_2, \beta_2, \gamma_2)$, expanding $(\alpha_1 x + \beta_1 y + \gamma_1)(\alpha_2 x + \beta_2 y + \gamma_2)$ gives

$$a = \alpha_1\alpha_2, \quad 2b = \alpha_1\beta_2 + \alpha_2\beta_1, \quad c = \beta_1\beta_2, \quad 2d = \alpha_1\gamma_2 + \alpha_2\gamma_1, \quad 2e = \beta_1\gamma_2 + \beta_2\gamma_1, \quad f = \gamma_1\gamma_2,$$

which are exactly the entries of $\frac{1}{2}(uw^T + wu^T)$:

$$M = \frac{1}{2}\begin{pmatrix} 2\alpha_1\alpha_2 & \alpha_1\beta_2 + \alpha_2\beta_1 & \alpha_1\gamma_2 + \alpha_2\gamma_1 \\ \alpha_1\beta_2 + \alpha_2\beta_1 & 2\beta_1\beta_2 & \beta_1\gamma_2 + \beta_2\gamma_1 \\ \alpha_1\gamma_2 + \alpha_2\gamma_1 & \beta_1\gamma_2 + \beta_2\gamma_1 & 2\gamma_1\gamma_2 \end{pmatrix}.$$

The converse also holds (a standard fact from linear algebra, not proved here): $\Delta = 0$ exactly when p is a product of two linear polynomials, possibly with complex coefficients. So the condition $\Delta \neq 0$ in Theorem 1 says that the conic is *non-degenerate*: a genuine ellipse, hyperbola or parabola, not a pair of lines or a single point.

Can a rigid motion remove the linear and constant terms? This question was raised in class. The answer is no when $\Delta \neq 0$. Without those terms the equation would be $ax^2 + 2bxy + cy^2 = 0$, whose matrix has last row $(0,0,0)$ and so $\Delta = 0$; but a rigid motion does not change Δ. More precisely:

- if $\delta \neq 0$, the linear terms can be removed (move the center to the origin), but the constant term is then $\Delta/\delta \neq 0$, by Proposition 3;
- if $\delta = 0$, not even all the linear terms can be removed: one remains, as in (3).

The substitution $x \mapsto x + \frac{b}{a}y$ used when completing the square does remove the xy term, but it is a shear, not a rigid motion: it changes the shape of the curve.

8 Two numbers are enough

Theorem 5. *Two conics with $\Delta \neq 0$ can be moved onto each other by a rigid motion if and only if they have the same values of f_1 and f_2.*

One direction is Theorem 1. For the other, the values of f_1, f_2 determine the normal form (2) or (3), and two conics with the same normal form are moved onto each other by going through it.

> Calculation
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> *Central case* ($\delta \neq 0$, equivalently $f_2 \neq 0$). Multiply the equation so that the constant term Δ/δ in (2) equals -1; then $\Delta = -\delta = -\lambda_1\lambda_2$, and
>
> $$f_2 = \frac{\lambda_1\lambda_2}{(\lambda_1\lambda_2)^{2/3}} = (\lambda_1\lambda_2)^{1/3}, \qquad f_1 = \frac{\lambda_1+\lambda_2}{(-\lambda_1\lambda_2)^{1/3}} = -\frac{\lambda_1+\lambda_2}{f_2}.$$
>
> So $\lambda_1\lambda_2 = f_2^3$ and $\lambda_1 + \lambda_2 = -f_1 f_2$: the numbers λ_1, λ_2 are the two roots of
>
> $$t^2 + f_1 f_2\, t + f_2^3 = 0,$$
>
> determined up to their order, and exchanging them is a rotation by $90°$.
> *Parabolic case* ($\delta = 0$, equivalently $f_2 = 0$). Multiply the equation so that $T = 1$ in (3); then $\Delta = -k^2$ and $f_1 = (-k^2)^{-1/3} = -|k|^{-2/3}$, which determines $|k|$. The sign of k is changed by the rotation by π.

The type of a conic. Since $\Delta^{2/3} > 0$, the sign of f_2 is the sign of δ, and the normal forms show what it means. For $\Delta \neq 0$:

- $f_2 > 0$ and $f_1 < 0$: an ellipse;
- $f_2 > 0$ and $f_1 > 0$: no real points (like $x^2 + y^2 + 1 = 0$);
- $f_2 < 0$: a hyperbola;
- $f_2 = 0$: a parabola (then always $f_1 < 0$).

For the standard equations, with $\alpha, \beta, q > 0$:

$$\begin{aligned}
&\text{ellipse } \frac{x^2}{\alpha^2} + \frac{y^2}{\beta^2} = 1: && f_1 = -\frac{\alpha^2+\beta^2}{(\alpha\beta)^{4/3}}, && f_2 = (\alpha\beta)^{-2/3},\\
&\text{hyperbola } \frac{x^2}{\alpha^2} - \frac{y^2}{\beta^2} = 1: && f_1 = \frac{\beta^2-\alpha^2}{(\alpha\beta)^{4/3}}, && f_2 = -(\alpha\beta)^{-2/3},\\
&\text{parabola } y^2 = 4qx: && f_1 = -(4q^2)^{-1/3}, && f_2 = 0.
\end{aligned}$$

For the ellipse, f_2 gives the product $\alpha\beta$ and then f_1 gives $\alpha^2 + \beta^2$, so the two half-axes are recovered, as Theorem 5 promises.

> Calculation
>
> *Type.* In (2) with $\delta = \lambda_1\lambda_2 > 0$, both λ_i have the sign of T, and there are real points exactly when the constant Δ/δ has the opposite sign, that is, when $T\Delta < 0$, that is, when $f_1 < 0$. With $\delta < 0$ the λ_i have opposite signs: a hyperbola. In the parabolic case $T\Delta = -T^2k^2 < 0$.
> *Ellipse.* Multiply by $\alpha^2\beta^2$: $\beta^2x^2 + \alpha^2y^2 - \alpha^2\beta^2 = 0$, so $T = \alpha^2 + \beta^2$, $\delta = \alpha^2\beta^2$, $\Delta = -\alpha^4\beta^4$.
> *Hyperbola.* $\beta^2x^2 - \alpha^2y^2 - \alpha^2\beta^2 = 0$, so $T = \beta^2 - \alpha^2$, $\delta = -\alpha^2\beta^2$, $\Delta = \alpha^4\beta^4$.
> *Parabola.* $y^2 - 4qx = 0$, so $c = 1$, $d = -2q$, all others 0: $T = 1$, $\delta = 0$, $\Delta = -4q^2$.

9 The geometric definitions agree

The sign of $\delta = ac - b^2$ can also be read off from the classical definitions of the three curves by distances.

Ellipse and hyperbola. Let F_1, F_2 be two points (the foci) at distance $2m$ from each other, and d_1, d_2 the distances from a point $P = (x, y)$ to them. The ellipse is where $d_1 + d_2 = L$, and the hyperbola where $|d_1 - d_2| = L$, for a fixed length L. Squaring twice turns either condition into the same polynomial equation,

$$L^4 - 2L^2\big(d_1^2 + d_2^2\big) + \big(d_1^2 - d_2^2\big)^2 = 0,$$

which is quadratic in x, y, with

$$\delta = ac - b^2 = 16L^2\big(L^2 - 4m^2\big).$$

For the ellipse $L = 2\alpha$ and $\alpha^2 = \beta^2 + m^2$, so $L^2 - 4m^2 = 4\beta^2 > 0$. For the hyperbola $L = 2\alpha$ and $m^2 = \alpha^2 + \beta^2$, so $L^2 - 4m^2 = -4\beta^2 < 0$. (Here α, β are the half-axes, as in the standard equations above.)

> Calculation
>
> *Squaring.* From $d_1 \pm d_2 = \pm L$: $d_1^2 + d_2^2 \pm 2d_1d_2 = L^2$, so $\pm 2d_1d_2 = L^2 - (d_1^2 + d_2^2)$. Squaring again,
>
> $$4d_1^2d_2^2 = L^4 - 2L^2(d_1^2 + d_2^2) + (d_1^2 + d_2^2)^2,$$
>
> and $(d_1^2 + d_2^2)^2 - 4d_1^2d_2^2 = (d_1^2 - d_2^2)^2$ gives the equation above. Every sign choice gives the same equation, which is why the ellipse and the hyperbola lead to the same formula.
> *The coefficients.* With $F_i = (h_i, k_i)$, $u = h_2 - h_1$, $w = k_2 - k_1$ (so $u^2 + w^2 = 4m^2$):
>
> $$d_1^2 + d_2^2 = 2x^2 + 2y^2 + (\text{linear}), \qquad d_1^2 - d_2^2 = 2ux + 2wy + (\text{constant}).$$
>
> So the quadratic part of the equation is $-4L^2(x^2 + y^2) + 4(ux + wy)^2$, that is,
>
> $$a = 4u^2 - 4L^2, \qquad b = 4uw, \qquad c = 4w^2 - 4L^2,$$
>
> and
>
> $$ac - b^2 = 16\big[(u^2 - L^2)(w^2 - L^2) - u^2w^2\big] = 16\big[L^4 - L^2(u^2 + w^2)\big] = 16L^2(L^2 - 4m^2).$$

Parabola. The parabola with focus (h, k) and directrix the line $rx + ty + s = 0$ is the set of points at equal distance from both:

$$\sqrt{(x - h)^2 + (y - k)^2} = \frac{|rx + ty + s|}{\sqrt{r^2 + t^2}}.$$

Its quadratic part is $t^2x^2 - 2rt\,xy + r^2y^2$, so $a = t^2$, $b = -rt$, $c = r^2$, and $\delta = r^2t^2 - r^2t^2 = 0$.

Calculation

Square and multiply by $r^2 + t^2$:

$$(r^2 + t^2)\big((x-h)^2 + (y-k)^2\big) - (rx + ty + s)^2 = 0.$$

The quadratic terms are $(r^2 + t^2)(x^2 + y^2) - (r^2x^2 + 2rt\,xy + t^2y^2) = t^2x^2 - 2rt\,xy + r^2y^2$.